Stakeholder Analysis

Stakeholder	Interests
1. Principal	• Simple system • Controlled release of marks
2. Lecture	• Create assignments • Provide extensions to deadlines • Grading and giving feedback
3. Head of Department	• Review marks before they are released • And also, to approve and release marks
4. Student Representative	• Fix a mistake and resubmit before the deadline • Receive written feedback
5. IT Officer	• Offline operation • Password and security • Good performance and maintainability

Conflicts and Resolution

1. Principal vs lecture

   The Principal says, “Students must not see any mark until it has been checked by the Head of Department.” While the Lecture says, ““I want students to get their marks as soon as I have marked the work”.

   Resolution: Marks remain hidden from students until the Head of Department has moderated and approved them.

2. Head of Department vs Student Representative

   The Head of Department says, “Late work should lose 10% of the available mark for each day it is late, for up to three days.” While the Student Representative says, “Late work should not simply be rejected.”

Resolution: Accept late work for up to three days and apply the specified penalty. After three days, the mark is zero.

3. Lecture vs Head of Department

   The Lecture says, “I want students to get their marks as soon as I have marked the work.” While the Head of Department says, “I need to see every mark in my department before it is released.”

   Resolution: Lecturer can complete marking, but the Head of Department must moderate and approve the mark before it becomes visible to the student.

Ambiguities

No.	Stakeholder statement	Ambiguity	Clarification needed
1	Principal: system must be "simple"	What exactly makes the system simple?	Define usability requirements, such as clear menus and straightforward workflows.
2	Lecturer: student may have a "good reason" for needing more time	What qualifies as a good reason, and who decides?	Define who can grant an extension and what information must be recorded.
3	Head of Department: 10% penalty "for each day it is late"	How is a partial day treated?	Define whether lateness is calculated by calendar day or another defined period.
4	Head of Department: late work is allowed "up to three days"	Does exactly three days late receive a penalty or zero?	Define the boundary explicitly: submissions within the permitted three-day period receive the applicable penalty; after that, zero.
5	Student Representative: resubmit "before the deadline"	How many times can a student resubmit?	Define whether multiple resubmissions are permitted and which submission is assessed.
6	IT Officer: system should "respond quickly"	What response time is considered quick?	Define a measurable response-time requirement.

7	IT Officer: "all 800 students"	Does this mean 800 registered students, or 800 simultaneous users?	Clarify the performance requirement and expected concurrent usage.
8	IT Officer: passwords must be "protected"	What level of password protection is required?	Define secure password storage and authentication requirements.

Software Requirements Specification

Functional Requirements

ID	Requirements	Priority	Verification
FR-01	The system should allow users to login using a unique username and password	Must	Test
FR-02	The system shall restrict functionality according to the user's assigned role	Must	Test
FR-03	The system should prevent unauthorized users from accessing protected functions	Must	Test
FR-04	The system should allow the administrator to create, update and deactivate user accounts	Must	Test
FR-05	The system should allow the administrator to create and manage departments and courses	Must	Test
FR-06	The system should allow the administrator to assign lectures to courses and enroll students in courses	Must	Test
FR-07	The system should allow the lecture to create assignments with a deadline and maximum mark	Must	Test
FR-08	The system should allow the lecture to grant individual students an extension and record the revised deadline	Must	Test
FR-09	The system should allow the lecture to mark submissions and provide written feedback	Must	Test
FR-10	The system should allow the lecture to view submissions for courses assigned to them	Must	Test
FR-11	The system should allow students to view their enrolled courses and available assignments	Must	Test
FR-12	The system should allow students to submit assignments before deadlines	Must	Test
FR-13	The system should record every change made to a mark	Must	Test
FR-14	The system shall provide clear confirmation when a student submits or resubmits an assignment.	Should	Test

Non-functional Requirements

ID	Requirement	Priority	Verification
NFR-01	The system should complete a login operation within 2 seconds when the database contains up to 800 student records	Must	Performance test
NFR-02	The system should support normal operations without requiring an internet connection	Must	System test
NFR-03	The system should never store user passwords in plaintext	Must	Security test
NFR-04	The system should handle invalid user input without terminating unexpectedly	Must	System test
NFR-05	The system should record 100% of changes made to student marks in the audit log	Must	Integration test
NFR-06	The system should enforce role-based access control to prevent unauthorized access to functions and records	Must	Security test
NFR-07	The system shall run unit and integration tests automatically through a continuous integration (CI) pipeline	Should	CI test

Use Case Diagram

